

Notes on Varieties of Semiring Solvers

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1 PRELIMINARIES

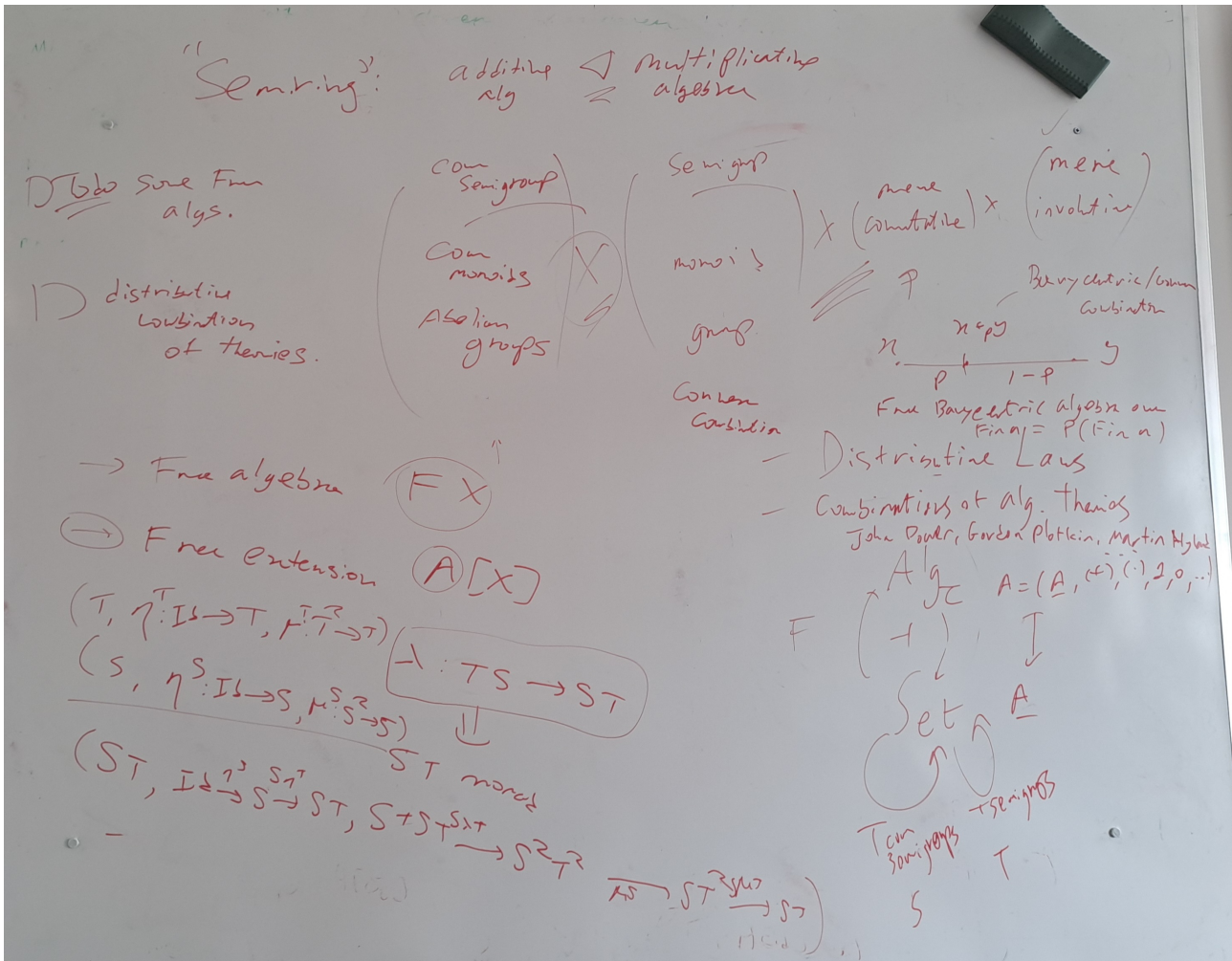


Figure 1: Blackboard at the start of the internship.

For convenience, we recall some definitions we'll be using.

1.1 Algebraic Structures

Definition 1 (Semigroup)

A *semigroup* is a set S equipped with an associative binary operation $\cdot : S \times S \rightarrow S$ such that for all $a, b, c \in S$:

- $a \cdot (b \cdot c) = (a \cdot b) \cdot c$ (associativity)

Definition 2 (Monoid)

A *monoid* is a set M equipped with an associative binary operation $\cdot : M \times M \rightarrow M$ and an identity element $e \in M$ such that for all $a, b, c \in M$:

- $a \cdot (b \cdot c) = (a \cdot b) \cdot c$ (associativity)
- $e \cdot a = a \cdot e = a$ (identity element)

Definition 3 (Group)

A *group* is a set G equipped with an associative binary operation $\cdot : G \times G \rightarrow G$, an identity element $e \in G$, and an inverse operation such that for all $a, b, c \in G$:

- $a \cdot (b \cdot c) = (a \cdot b) \cdot c$ (associativity)
- $e \cdot a = a \cdot e = a$ (identity element)
- For each $a \in G$, there exists an element $a^{-1} \in G$ such that $a \cdot a^{-1} = a^{-1} \cdot a = e$ (inverse element)

A group is called *abelian* if its operation is commutative, i.e., for all $a, b \in G$, $a \cdot b = b \cdot a$.

Definition 4 (Semiring)

A *semiring* is a set R equipped with two binary operations: addition $+$ and multiplication $\cdot$, such that:

- $(R, +)$ is a commutative monoid with identity element 0.
- $(R, \cdot)$ is a monoid with identity element 1.
- Multiplication distributes over addition from both sides: for all $a, b, c \in R$,

$$\begin{aligned} a \cdot (b + c) &= a \cdot b + a \cdot c, \\ (a + b) \cdot c &= a \cdot c + b \cdot c. \end{aligned}$$

- The additive identity 0 is an absorbing element for multiplication: for all $a \in R$, $0 \cdot a = a \cdot 0 = 0$.

Definition 5 (Operad)

An *operad* $\mathcal{O}$ consists of:

- A collection of sets $\{\mathcal{O}_n\}_{n \geq 0}$, where $\mathcal{O}_n$ is the set of n -ary operations,
- A composition function $\gamma : \mathcal{O}_n \times \mathcal{O}_{k_1} \times \cdots \times \mathcal{O}_{k_n} \rightarrow \mathcal{O}_{k_1 + \cdots + k_n}$,
- An identity element $e \in \mathcal{O}_1$,

satisfying the following axioms:

- Associativity: For all $f \in \mathcal{O}_n$, $g_i \in \mathcal{O}_{k_i}$, and $h_{ij} \in \mathcal{O}_{m_{ij}}$, we have

$$\gamma(f; g_1, \dots, g_n) = f(g_1(h_{11}, \dots, h_{1k_1}), \dots, g_n(h_{n1}, \dots, h_{nk_n})).$$

- Identity: For all $f \in \mathcal{O}_n$, we have

$$f(e, \dots, e) = f.$$

Definition 6 (Algebraic Theory)

An *algebraic theory* $\mathcal{T}$ consists of:

- A collection of sets $\{\mathcal{T}_n\}_{n \geq 0}$, where $\mathcal{T}_n$ is the set of n -ary operations,
- A collection of equations between terms built from these operations,

such that the operations and equations define a class of algebras (models) that satisfy the given equations.

Example (Operad from an Algebraic Theory)

Given an algebraic theory $\mathcal{T}$, we can construct an operad $\mathcal{O}^{\mathcal{T}}$ where:

- $\mathcal{O}_n^{\mathcal{T}}$ consists of the n -ary operations in $\mathcal{T}$,
- The composition function γ is defined by substituting operations according to the equations of the theory,
- The identity element e corresponds to the identity operation in the theory.

1.2 Categories**Definition 7 (Category)**

A *category* $\mathcal{C}$ consists of:

- A class of objects $\text{Ob}(\mathcal{C})$,
- For each pair of objects $a, b \in \text{Ob}(\mathcal{C})$, a set of morphisms $\text{Hom}_{\mathcal{C}}(a, b)$,
- For each object $a \in \text{Ob}(\mathcal{C})$, an identity morphism $\text{id}_a \in \text{Hom}_{\mathcal{C}}(a, a)$,
- A composition operation $\circ$ such that for morphisms $f : a \rightarrow b$ and $g : b \rightarrow c$, there is a morphism $g \circ f : a \rightarrow c$,

satisfying the following axioms:

- **Associativity:** For morphisms $f : a \rightarrow b$, $g : b \rightarrow c$, and $h : c \rightarrow d$, we have $h \circ (g \circ f) = (h \circ g) \circ f$.
- **Identity:** For any morphism $f : a \rightarrow b$, we have $\text{id}_b \circ f = f = f \circ \text{id}_a$.

Example (Set Category)

The category **Set** has sets as objects and functions as morphisms. The identity morphism for a set a is the identity function $\text{id}_a : a \rightarrow a$ defined by $\text{id}_a(x) = x$ for all $x \in a$. Composition of morphisms is given by function composition.

Definition 8 (Product)

Given two objects a and b in a category $\mathcal{C}$, a *product* of a and b is an object $a \times b$ together with two morphisms $\pi_a : a \times b \rightarrow a$ and $\pi_b : a \times b \rightarrow b$ such that for any object c and morphisms $f : c \rightarrow a$ and $g : c \rightarrow b$, there exists a unique morphism $\langle f, g \rangle : c \rightarrow a \times b$ making the following diagram commute:

$$\begin{array}{ccccc}
 & & c & & \\
 & f \swarrow & \downarrow \langle f, g \rangle & \searrow g & \\
 a & \xleftarrow{\pi_a} & a \times b & \xrightarrow{\pi_b} & b
 \end{array}$$

Definition 9 (Coproduct)

Given two objects a and b in a category $\mathcal{C}$, a *coproduct* of a and b is an object $a + b$ together with two morphisms $\iota_a : a \rightarrow a + b$ and $\iota_b : b \rightarrow a + b$ such that for any object c and morphisms $f : a \rightarrow c$ and $g : b \rightarrow c$, there exists a unique morphism $[f, g] : a + b \rightarrow c$ making the following diagram commute:

$$\begin{array}{ccccc} a & \xrightarrow{\iota_a} & a + b & \xleftarrow{\iota_b} & b \\ & \searrow f & \downarrow [f, g] & \swarrow g & \\ & & c & & \end{array}$$

Definition 10 (Monoidal Category)

A *monoidal category* is a category $\mathcal{C}$ equipped with a bifunctor $\otimes : \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$, an object I called the unit object, and natural isomorphisms (called associators and unitors) that satisfy certain coherence conditions. Specifically, for all objects a, b, c in $\mathcal{C}$, there are natural isomorphisms:

- Associator: $\alpha_{a,b,c} : (a \otimes b) \otimes c \rightarrow a \otimes (b \otimes c)$,
- Left unitor: $\lambda_a : I \otimes a \rightarrow a$,
- Right unitor: $\rho_a : a \otimes I \rightarrow a$,

such that the following diagrams commute:

$$\begin{array}{ccc} ((a \otimes b) \otimes c) \otimes d \xrightarrow{\alpha_{a,b,c} \otimes 1_d} (a \otimes (b \otimes c)) \otimes d \xrightarrow{\alpha_{a,b \otimes c,d}} a \otimes ((b \otimes c) \otimes d) \\ \alpha_{a,b,c \otimes d} \uparrow & & 1_a \otimes \alpha_{b,c,d} \uparrow \\ (a \otimes b) \otimes (c \otimes d) & & a \otimes (b \otimes (c \otimes d)) \end{array}$$

$$(I \otimes a) \otimes b \xrightarrow{\alpha_{I,a,b}} I \otimes (a \otimes b) \xleftarrow{\rho_a} a = a \otimes I \xrightarrow{\lambda_a} a \otimes I \xrightarrow{\alpha_{a,I,I}} (a \otimes I) \otimes I$$

Definition 11 (Functor)

Given two categories $\mathcal{C}$ and $\mathcal{D}$, a *functor* $F : \mathcal{C} \rightarrow \mathcal{D}$ consists of:

- A mapping $F : \text{Ob}(\mathcal{C}) \rightarrow \text{Ob}(\mathcal{D})$ that assigns to each object a in $\mathcal{C}$ an object $F(a)$ in $\mathcal{D}$,
- For each pair of objects $a, b \in \text{Ob}(\mathcal{C})$, a mapping $F : \text{Hom}_{\mathcal{C}}(a, b) \rightarrow \text{Hom}_{\mathcal{D}}(F(a), F(b))$ that assigns to each morphism $f : a \rightarrow b$ in $\mathcal{C}$ a morphism $F(f) : F(a) \rightarrow F(b)$ in $\mathcal{D}$,

such that for all objects a, b, c in $\mathcal{C}$ and morphisms $f : a \rightarrow b$, $g : b \rightarrow c$, the following conditions hold:

- $F(g \circ f) = F(g) \circ F(f)$ (preservation of composition),
- $F(\text{id}_a) = \text{id}_{F(a)}$ (preservation of identity).

For a category $\mathcal{C}$, the *identity functor* $\text{Id}_{\mathcal{C}} : \mathcal{C} \rightarrow \mathcal{C}$ is defined by $\text{Id}_{\mathcal{C}}(a) = a$ for all objects a and $\text{Id}_{\mathcal{C}}(f) = f$ for all morphisms f .

- A functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is *faithful* if, for every pair of objects $a, b \in \text{Ob}(\mathcal{C})$, the map $F : \text{Hom}_{\mathcal{C}}(a, b) \rightarrow \text{Hom}_{\mathcal{D}}(F(a), F(b))$ is injective.
- A functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is *full* if, for every pair of objects $a, b \in \text{Ob}(\mathcal{C})$, the map $F : \text{Hom}_{\mathcal{C}}(a, b) \rightarrow \text{Hom}_{\mathcal{D}}(F(a), F(b))$ is surjective.
- A functor that is both full and faithful is called *fully faithful*.

Notation: We often denote the image of an object a under a functor F as Fa instead of $F(a)$ and the image of a morphism $f : a \rightarrow b$ as $Ff : Fa \rightarrow Fb$.

Example (Diagonal functor)

Given a category $\mathcal{C}$, the *diagonal functor* $\Delta : \mathcal{C} \rightarrow \mathcal{C} \times \mathcal{C}$ is defined by $\Delta(a) = (a, a)$ for each object a in $\mathcal{C}$ and $\Delta(f) = (f, f)$ for each morphism $f : a \rightarrow b$ in $\mathcal{C}$.

Definition 12 (Natural Transformation)

Given two functors $F, G : \mathcal{C} \rightarrow \mathcal{D}$ between categories $\mathcal{C}$ and $\mathcal{D}$, a *natural transformation* $\eta : F \rightarrow G$ consists of a family of morphisms $\{\eta_a : Fa \rightarrow Ga\}_{a \in \text{Ob}(\mathcal{C})}$ such that for every morphism $f : a \rightarrow b$ in $\mathcal{C}$, the following diagram commutes:

$$\begin{array}{ccc} Fa & \xrightarrow{Ff} & Fb \\ \eta_a \downarrow & & \downarrow \eta_b \\ Ga & \xrightarrow{Gf} & Gb \end{array}$$

Definition 13 (Monad)

A *monad* on a category $\mathcal{C}$ is a triple (T, η, μ) where:

- $T : \mathcal{C} \rightarrow \mathcal{C}$ is a functor,
- $\eta : \text{Id}_{\mathcal{C}} \rightarrow T$ is a natural transformation called the unit,
- $\mu : T^2 \rightarrow T$ is a natural transformation called the multiplication,

such that the following diagrams commute:

$$\begin{array}{ccc} T & \xrightarrow{T\eta} & T^2 \\ \downarrow \eta T & & \downarrow \mu \\ T^2 & \xrightarrow{\mu} & T \end{array} \quad \begin{array}{ccc} T^3 & \xrightarrow{T\mu} & T^2 \\ \downarrow \mu T & & \downarrow \mu \\ T^2 & \xrightarrow{\mu} & T \end{array}$$

Definition 14 (Adjunction)

Given two categories $\mathcal{C}$ and $\mathcal{D}$, a functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is a *left adjoint* to a functor $G : \mathcal{D} \rightarrow \mathcal{C}$ (and G is a *right adjoint* to F) if there is a natural isomorphism:

$$\text{Hom}_{\mathcal{D}}(Fa, b) \cong \text{Hom}_{\mathcal{C}}(a, Gb)$$

for all objects a in $\mathcal{C}$ and b in $\mathcal{D}$. This means that for every morphism $f : Fa \rightarrow b$ in $\mathcal{D}$, there is a unique morphism $g : a \rightarrow Gb$ in $\mathcal{C}$ such that the following diagram commutes:

$$\begin{array}{ccc} Fa & \xrightarrow{f} & b \\ \uparrow F & & \downarrow G \\ a & \xrightarrow{g} & Gb \end{array}$$

We often denote this adjunction by $F \dashv G$.

Example (Monad from an Adjunction)

Given an adjunction $F \dashv G$ between categories $\mathcal{C}$ and $\mathcal{D}$, we can construct a monad on $\mathcal{C}$ as

follows:

- The functor $T : \mathcal{C} \rightarrow \mathcal{C}$ is defined by $T = GF$.
- The unit $\eta : \text{Id}_{\mathcal{C}} \rightarrow T$ is given by the natural transformation that assigns to each object a in $\mathcal{C}$ the morphism $\eta_a : a \rightarrow GFa$ corresponding to the identity morphism on Fa under the adjunction.
- The multiplication $\mu : T^2 \rightarrow T$ is given by the natural transformation that assigns to each object a in $\mathcal{C}$ the morphism $\mu_a : GF GFa \rightarrow GFa$ corresponding to the composition of morphisms in $\mathcal{D}$ under the adjunction.

There are actually several different but equivalent ways to define adjunctions. They are linked thanks to the Yoneda lemma.

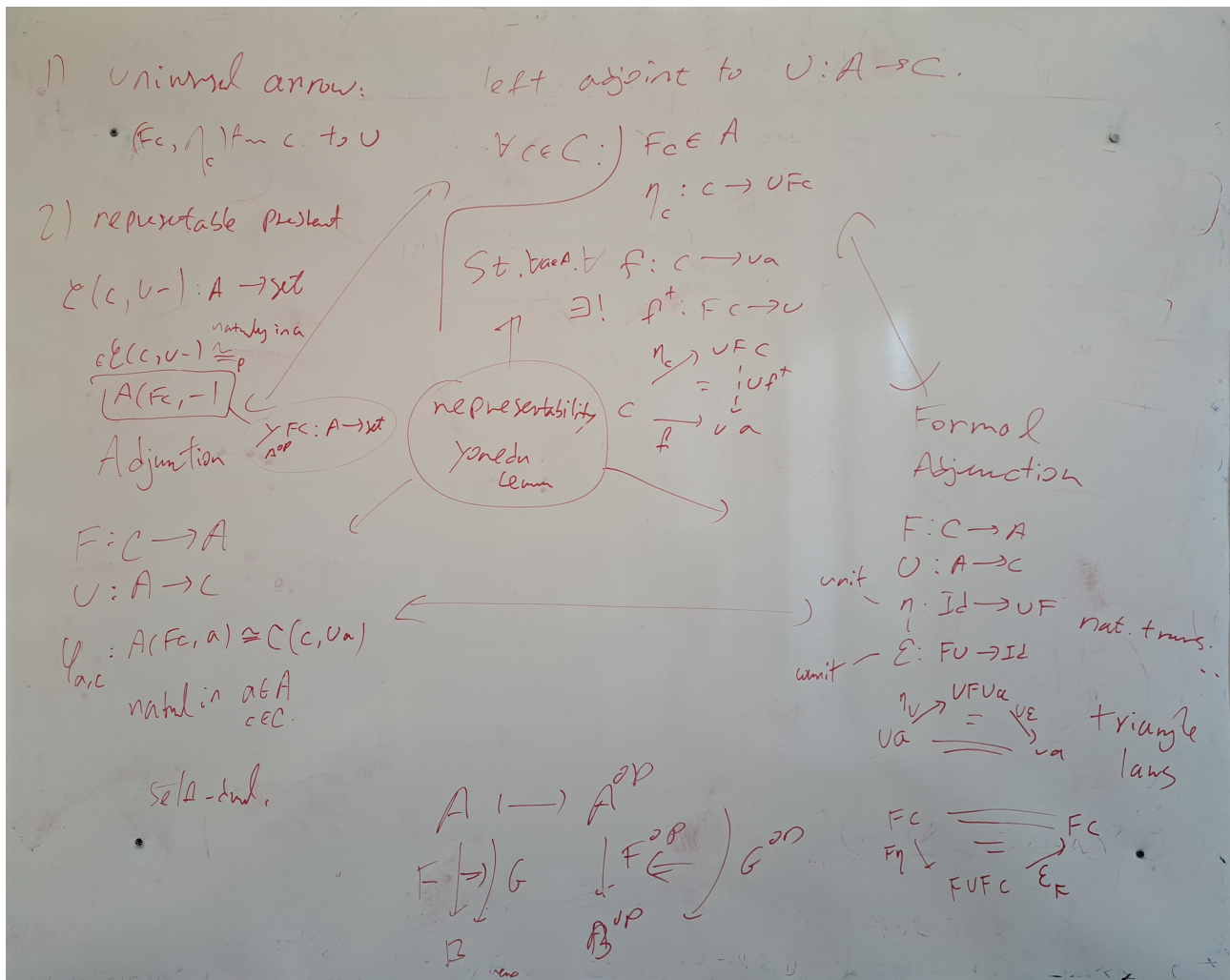


Figure 2: Different equivalent definitions of adjunctions.

One important definition is that of the universal arrow.

Definition 15 (Universal arrow)

Let $U : A \rightarrow C$ be a functor, and let $c \in C$.

An *universal arrow* from c to U is a pair (Fc, η_c) where:

- $Fc \in A$
- $\eta_c : c \rightarrow UFc$ is a morphism in C

such that for every $a \in A$, and every $f : c \rightarrow Ua$ in C , there exists a unique morphism $f^\dagger : Fc \rightarrow a$ in A such that the following diagram commutes:

$$\begin{array}{ccc} c & \xrightarrow{\eta_c} & UFc \\ & \searrow f & \downarrow Uf^\dagger \\ & & Ua \end{array} \quad \begin{array}{ccc} Fc & & \\ \downarrow f^\dagger & & \\ a & & \end{array}$$

We can dualize this definition to get the notion of a universal arrow from U to c .

A universal arrow from U to c is a pair (Fc, η_c) where:

- $Fc \in A$
- $\eta_c : UFc \rightarrow c$ is a morphism in C

such that for every $a \in A$, and every $f : Ua \rightarrow c$ in C , there exists a unique morphism $f^\dagger : a \rightarrow Fc$ in A such that the following diagram commutes:

$$\begin{array}{ccc} c & \xleftarrow{\eta_c} & UFc \\ & \swarrow f & \uparrow Uf^\dagger \\ & & Ua \end{array} \quad \begin{array}{ccc} Fc & & \\ \uparrow f^\dagger & & \\ a & & \end{array}$$

Example (Adjunction from Universal Arrows)

Given a functor $U : A \rightarrow C$ such that for every object $c \in C$, there exists a universal arrow from c to U , we can construct an adjunction $F \dashv U$ where:

- The functor $F : C \rightarrow A$ is defined by sending each object c in C to the object Fc in A that appears in the universal arrow from c to U , and sending each morphism $f : c \rightarrow c'$ in C to the unique morphism $Ff : Fc \rightarrow Fc'$ in A that makes the corresponding diagram commute.
- The unit $\eta : \text{Id}_C \rightarrow UF$ is given by the family of morphisms $\{\eta_c : c \rightarrow UFc\}_{c \in C}$ that are part of the universal arrows.

Universal arrows have more uses than just defining adjunctions. Here are other examples.

Example (Defining the coproduct with an universal arrow)

Let A be a category, and let $a, b \in A$. A *coproduct* of a and b is a universal arrow from (a, b) to the diagonal functor $\Delta : A \rightarrow A \times A$, where $\Delta(c) = (c, c)$ for any object $c \in A$.

$$\begin{array}{ccc} a & \xrightarrow{\iota_a} & a + b \xleftarrow{\iota_b} b \\ & \searrow f & \downarrow [f, g] \\ & & c \end{array} \quad \sim \quad \begin{array}{ccc} (a, b) & \xrightarrow{\eta_{(a,b)}} & \Delta(a + b) \\ & \searrow (f, g) & \downarrow \Delta[f, g] \\ & & \Delta c \end{array}$$

We have $\eta_{(a,b)} = (\iota_a, \iota_b)$.

Example (Defining the product with an universal arrow)

Let A be a category, and let $a, b \in A$. A *product* of a and b is a universal arrow from the diagonal functor $\Delta : A \rightarrow A \times A$ to (a, b) .

$$\begin{array}{ccc}
 & c & \\
 f \swarrow & \downarrow (f,g) & \searrow g \\
 a & \xleftarrow{\pi_a} a \times b \xrightarrow{\pi_b} & b
 \end{array}
 \quad \sim \quad
 \begin{array}{ccc}
 \Delta c & \xrightarrow{\eta_{(a,b)}} & \Delta(a \times b) \\
 (f,g) \searrow & & \downarrow (\pi_a, \pi_b) \\
 & & (a, b)
 \end{array}$$

We have $\eta_{(a,b)} = \langle f, g \rangle$.

1.3 2-categories**Definition 16 (2-category)**

A *2-category* C consists of:

- Cellular structure:
 - A collection C_0 of 0-cells $a, b, c, \dots$ (objects),
 - For each pair of 0-cells a, b , a collection $C_1(a, b)$ of 1-cells (morphisms) from a to b ,
 - For each pair of 1-cells $f, g \in C_1(a, b)$, a collection $C_2(f, g)$ of 2-cells (2-morphisms) from f to g
- Identities:
 - A 1-cell $\text{id}_a : a \rightarrow a$ for each 0-cell $a \in C_0$ (identity 1-cell),
 - A 2-cell $\text{id}_f : f \Rightarrow f$ for each 1-cell $f \in C_1(a, b)$ (identity 2-cell)
- Composition:
 - A composition operation on 1-cells $g \circ f$ for every pair of composable 1-cells $f : a \rightarrow b$ and $g : b \rightarrow c$,
 - A vertical composition operation on 2-cells $\beta \circ \alpha : f \Rightarrow h$ for every pair of composable 2-cells $\alpha : f \Rightarrow g$ and $\beta : g \Rightarrow h$,
 - A horizontal composition operation on 2-cells $\beta \bullet \alpha : g \circ f \Rightarrow g' \circ f'$ for every pair of composable 2-cells $\alpha : f \Rightarrow f'$ and $\beta : g \Rightarrow g'$

such that

- 1-cell compositions are associative and identities are neutral
- 2-cell vertical compositions are associative and identities are neutral
- 2-cell horizontal compositions are associative and identities are neutral, and composition of 2-cell identities preserves 1-cell composition
- the interchange law holds: for any composable 2-cells $\alpha, \alpha', \beta, \beta'$, we have

$$(\beta' \circ \beta) \bullet (\alpha' \circ \alpha) = (\beta' \bullet \alpha') \circ (\beta \bullet \alpha)$$

Example (2-category of Categories)

The 2-category **Cat** has:

- 0-cells: categories,
- 1-cells: functors between categories,
- 2-cells: natural transformations between functors.
- pointwise composition of functors as 1-cell composition,
- pointwise composition of natural transformations as vertical composition,
- horizontal composition of natural transformations defined by the composition of functors.

Definition 17 (Adjunction in a 2-category)

Given a 2-category C , an *adjunction* between 0-cells a and b in C consists of:

- Two 1-cells $f : a \rightarrow b$ and $g : b \rightarrow a$,
- Two 2-cells $\eta : \text{id}_a \Rightarrow g \circ f$ (unit) and $\varepsilon : f \circ g \Rightarrow \text{id}_b$ (counit),

such that the following triangle identities hold:

$$(f \bullet \eta) \circ (\varepsilon \bullet f) = \text{id}_f \quad (\eta \bullet g) \circ (g \bullet \varepsilon) = \text{id}_g$$

1.4 Multicategories

Definition 18 (Multicategory)

A multicategory $\mathcal{U}$ consists of:

- A class of objects $\text{Ob}(\mathcal{U})$,
- For each finite (possibly empty) sequence of objects $(a_1, a_2, \dots, a_n)$ and an object b , a set of morphisms $\mathcal{U}(a_1, a_2, \dots, a_n; b)$,
- For each object a , an identity morphism $\text{id}_a : (a) \rightarrow a$,
- A composition operation that allows for the composition of morphisms in a way that generalizes the composition in a category, satisfying appropriate associativity and identity axioms.

We denote by **Multicat** the set of all multicategories.

Example (Category with Finite Products as a Multicategory)

Given a category $\mathcal{C}$ with finite products, we can construct a multicategory $\mathcal{U}$ where:

- The objects of $\mathcal{U}$ are the same as the objects of $\mathcal{C}$,
- The morphisms $\mathcal{U}(a_1, a_2, \dots, a_n; b)$ are given by the morphisms in $\mathcal{C}$ from the product $a_1 \times a_2 \times \dots \times a_n$ to b ,
- The identities rely on the canonical isomorphism

$$\text{id}_a := \prod_{i=1}^n a \cong a$$

- The composition is defined using the universal property of products and the composition in $\mathcal{C}$.

Example (Monoidal Category as a Multicategory)

A monoidal category $(\mathcal{C}, \otimes, I)$ can be viewed as a multicategory where:

- The objects of the multicategory are the same as the objects of $\mathcal{C}$,
- The morphisms $\mathcal{U}(a_1, a_2, \dots, a_n; b)$ are given by the morphisms in $\mathcal{C}$ from $a_1 \otimes a_2 \otimes \dots \otimes a_n$ to b (by nesting the monoidal product)

Definition 19 (Multifunctor)

A *multi-functor* $F : \mathcal{U} \rightarrow \mathcal{V}$ between multicategories $\mathcal{U}$ and $\mathcal{V}$ consists of two mappings:

- A mapping $F : \text{Ob}(\mathcal{U}) \rightarrow \text{Ob}(\mathcal{V})$ that assigns to each object a in $\mathcal{U}$ an object Fa in $\mathcal{V}$,
- A mapping $F : \mathcal{U}(a_1, a_2, \dots, a_n; b) \rightarrow \mathcal{V}(Fa_1, Fa_2, \dots, Fa_n; Fb)$ that assigns to each morphism $f : (a_1, a_2, \dots, a_n) \rightarrow b$ in $\mathcal{U}$ a morphism $Ff : (Fa_1, Fa_2, \dots, Fa_n) \rightarrow Fb$ in $\mathcal{V}$,

such that F preserves:

- Identities: For each object a in $\mathcal{U}$, $F\text{id}_a = \text{id}_{Fa}$,
- Composition: For morphisms $f : (a_1, a_2, \dots, a_n) \rightarrow b$ and $g_i : (a_{i1}, a_{i2}, \dots, a_{ik_i}) \rightarrow a_i$ in $\mathcal{U}$, we have

$$F(f \circ (g_1, g_2, \dots, g_n)) = Ff \circ (Fg_1, Fg_2, \dots, Fg_n).$$

Definition 20 (Multi-natural transformation)

Given two multi-functors $F, G : \mathcal{U} \rightarrow \mathcal{V}$ between multicategories $\mathcal{U}$ and $\mathcal{V}$, a *multi-natural transformation* $\eta : F \rightarrow G$ consists of a family of morphisms $\{\eta_a : Fa \rightarrow Ga\}_{a \in \text{Ob}(\mathcal{U})}$ such that for every morphism $f : (a_1, a_2, \dots, a_n) \rightarrow b$ in $\mathcal{U}$, the following diagram commutes:

$$\begin{array}{ccc} (Fa_1, Fa_2, \dots, Fa_n) & \xrightarrow{Ff} & Fb \\ (\eta_{a_1}, \eta_{a_2}, \dots, \eta_{a_n}) \downarrow & & \downarrow \eta_b \\ (Ga_1, Ga_2, \dots, Ga_n) & \xrightarrow{Gf} & Gb \end{array}$$

Definition 21 (Model)

Let $\mathcal{O}$ be an operad, and $\mathcal{U}$ be a multicategory. A *model* of $\mathcal{O}$ in $\mathcal{U}$ is a multi-functor $\mathcal{A} : \mathcal{O} \rightarrow \mathcal{U}$ that preserves the operadic structure, i.e., it respects the composition and identity of the operad.

Definition 22

Homomorphism of Models A *homomorphism* $h : \mathcal{A} \rightarrow \mathcal{B}$ between $\mathcal{O}$ -models in $\mathcal{U}$ consists of a morphism $\underline{h} : (\underline{\mathcal{A}}) \rightarrow (\underline{\mathcal{B}})$ such that for all $f \in \mathcal{O}_n$,

$$\underline{h} \circ (\mathcal{A}[[f]]) = \mathcal{B}[[f]](\underline{h}, \dots, \underline{h})$$

Example (Category of Models)

Given an operad $\mathcal{O}$ and a multicategory $\mathcal{U}$, we can construct a category $\mathbf{Mod}(\mathcal{O}, \mathcal{U})$ where:

- The objects are the models of $\mathcal{O}$ in $\mathcal{U}$,
- The morphisms are the homomorphisms between these models.

2 SOME FREE ALGEBRAS

2.1 Definitions

Definition 23 (Finitary signature)

A *finitary signature* $\Sigma = (\text{op } \Sigma, \text{arity})$ consists of a set $\text{op } \Sigma$ of *operators*, and an assignment $\text{arity} : \text{op } \Sigma \rightarrow \mathbb{N}$ associating to each operator in $\text{op } \Sigma$ its *arity*.

Definition 24 (Algebra)

An *algebra* $A = (\underline{A}, A[[_]])$ over a finitary signature Σ consists of:

- A set $\underline{A}$ called the *carrier* of A ,
- For each operator $f \in \text{op } \Sigma$ of arity n , a function $A[[f]] : \underline{A}^n \rightarrow \underline{A}$ called the *interpretation* of f in A .

Definition 25 (Terms)

Given a set X of variables and a finitary signature Σ , the set of *terms* $\text{Term}_\Sigma(X)$ is defined inductively as follows:

- Every variable $x \in X$ is a term,
- If $f \in \text{op } \Sigma$ is an operator of arity n and $t_1, t_2, \dots, t_n$ are terms, then $f(t_1, t_2, \dots, t_n)$ is a term.

Remark : The definition of algebraic theory seen before can be reformulated as a finitary signature together with a set of equations between terms built from the operators in the signature.

Definition 26 (Environment and Interpretation)

An *environment* ρ for a context X in an algebra A is a function $\rho : X \rightarrow \underline{A}$ that assigns to each variable in X an element of the carrier of A .

Given a term $t \in \text{Term}_\Sigma(X)$, denoted $X \vdash t$, the algebra A gives an *interpretation* function $A[[t]] : \underline{A}^X \rightarrow \underline{A}$. Given an environment $\rho : X \rightarrow \underline{A}$, we can evaluate the term t in A under the environment ρ , denoted $A[[t]](\rho)$, by recursively substituting the variables in t with their corresponding values in $\underline{A}$ according to ρ and applying the interpretations of the operators as defined in A .

Definition 27 (Valid equations)

An equation $t = t'$ between terms in $\text{Term}_\Sigma(X)$ is *valid* in an algebra A if for every environment $\rho : X \rightarrow \underline{A}$, we have $A[[t]](\rho) = A[[t']](\rho)$.

Definition 28 (Presentation and Models)

A *presentation* $\mathcal{T} = (\Sigma_{\mathcal{T}}, \mathcal{T}_{ax})$ consists of a finitary signature $\Sigma_{\mathcal{T}}$ and a set $\mathcal{T}_{ax}$ of *axioms*, which are equations between terms in $\text{Term}_{\Sigma_{\mathcal{T}}}(X)$ for some set of variables X .

A $\mathcal{T}$ -model A is a $\Sigma_{\mathcal{T}}$ -algebra such that all equations in $\mathcal{T}_{ax}$ are valid in A .

A $\mathcal{T}$ -model A over a context X is a $\mathcal{T}$ -model A together with an environment $\rho : X \rightarrow \underline{A}$.

Definition 29 (Homomorphisms)

Let $A = (\underline{A}, A[[\]])$ and $B = (\underline{B}, B[[\]])$ be algebras over the same signature Σ .

A *homomorphism* of Σ -algebras from A to B is a function $h : \underline{A} \rightarrow \underline{B}$ such that for every operator $f \in \text{op } \Sigma$ of arity n , we have:

$$B[[f]](h(a_1), \dots, h(a_n)) = h(A[[f]](a_1, \dots, a_n))$$

for all $a_1, \dots, a_n \in \underline{A}$.

A *homomorphism* of $\mathcal{T}$ -models is a homomorphism of $\Sigma_{\mathcal{T}}$ -algebras between the underlying algebras of the models.

Definition 30 (Morphism)

A *morphism* $h : A \rightarrow B$ of $\mathcal{T}$ -models over X is a $\mathcal{T}$ -model homomorphism that makes the diagram commute:

$$\begin{array}{ccc} X & \xrightarrow{\rho_A} & \underline{A} \\ & \searrow \rho_B & \downarrow h \\ & & \underline{B} \end{array}$$

Definition 31 (Free model (or free algebra))

Let $\mathcal{T}$ be a presentation and X be a set of variables.

A *free $\mathcal{T}$ -model* over X is a $\mathcal{T}$ -model $F(X)$ such that for any $\mathcal{T}$ -model A over X , there exists a unique morphism of $\mathcal{T}$ -models from $F(X)$ to A .

The existence-and-uniqueness property is called the *universal property* of free $\mathcal{T}$ -models over X .

Informally:

- A signature gives defines the operators and constants (arity 0 operators).
- An algebra gives a concrete interpretation of the signature.
- A presentation defines axioms on top of the signature, for instance the axioms of a group, or of a semiring, etc.
- A model of a presentation is an algebra that satisfies the axioms of the presentation.
- A free algebra is the most general model of a presentation.

We can give an equivalent definition for free algebras using universal arrows.

Proposition 32

Let $\mathcal{T}$ be a presentation, and X be a set. Let $\mathcal{T}\text{-}\mathbf{Mod}$ be the category of $\mathcal{T}$ -models over X . Let $U : \mathcal{T}\text{-}\mathbf{Mod} \rightarrow \mathbf{Set}$ be the forgetful functor that sends a $\mathcal{T}$ -model to its underlying set.

Then, a free $\mathcal{T}$ -model over X is a universal arrow from X to U .

Proof

Let $F(X)$ be a free $\mathcal{T}$ -model over X . We want to show that $(F(X), \eta_X)$ is a universal arrow from X to U , where $\eta_X : X \rightarrow U(F(X))$ is the function that sends each variable in X to its interpretation in $F(X)$.

Let A be a $\mathcal{T}$ -model over X , and let $\rho_A : X \rightarrow U(A)$ be the environment of A . Since $F(X)$ is a free $\mathcal{T}$ -model over X , there exists a unique morphism of $\mathcal{T}$ -models $h : F(X) \rightarrow A$ such that the following diagram commutes:

$$\begin{array}{ccc} X & \xrightarrow{\eta_X} & U(F(X)) \\ & \searrow \rho_A & \downarrow U(h) \\ & & U(A) \end{array}$$

This shows that $(F(X), \eta_X)$ satisfies the universal property of being a universal arrow from X to U .

Conversely, if $(F(X), \eta_X)$ is a universal arrow from X to U , then for any $\mathcal{T}$ -model A over X , there exists a unique morphism of $\mathcal{T}$ -models from $F(X)$ to A , which means that $F(X)$ is a free $\mathcal{T}$ -model over X . Therefore, the two definitions are equivalent. $\square$

2.2 Constructing some free algebras

Free commutative semigroups

Let's try to construct the free algebra for the presentation of commutative semigroups. We cannot rely on the order of the terms, nor the order of applying the operations.

Let X be a set. We consider the theory of commutative semigroups **CSG**, with a signature Σ with a single operator $(+)$ of arity 2, and with the two axioms

$$\begin{aligned} a, b &\vdash a + b = b + a \\ a, b, c &\vdash a + (b + c) = (a + b) + c \end{aligned}$$

We pose

$$SX = \{f : X \rightarrow \mathbb{N} \mid f \text{ has finite non-empty support}\}$$

For $f, g \in SX$, we pose

$$f + g = \lambda x. f(x) + g(x) \quad x \in X$$

We define $\eta_{\mathbf{CSG}} : X \rightarrow SX$ for all $x \in X$ by

$$\eta_{\mathbf{CSG}}(x) = \lambda y. \begin{cases} 1 & \text{if } x = y \\ 0 & \text{otherwise} \end{cases}$$

We pose $F_{\mathbf{CSG}} = (SX, (+))$ where $+$ is the operator defined above.

We want to show that $F_{\mathbf{CSG}}$ is a free **CSG**-model over X .

▷ Let's first check that $F_{\mathbf{CSG}}X$ is indeed a **CSG**-model. We need to check that the axioms of **CSG** are valid in $F_{\mathbf{CSG}}X$. This is straightforward and comes down to the commutativity and associativity of addition in $\mathbb{N}$.

▷ Let A be a **CSG**-model over X with environment $\varphi_A : X \rightarrow U(A)$. We want to show¹ that there exists a unique morphism h of **CSG**-models from $F_{\mathbf{CSG}}X$ to A such that the following diagram commutes:

$$\begin{array}{ccc} X & \xrightarrow{\eta_{\mathbf{CSG}}} & SX \\ & \searrow \varphi_A & \downarrow h \\ & & UA \end{array}$$

¹I think this is it ?

with U being the forgetful functor from **CSG**-models to **Set**.

We define $\Sigma\varphi_A : SX \rightarrow \underline{A}$ for all $f \in SX$ by

$$\Sigma\varphi_A(f) = \sum_{x \in X} f \ x \cdot \varphi_A(x)$$

We want to show that $\Sigma\varphi_A$ is the unique morphism h of **CSG**-models from $F_{\mathbf{CSG}}X$ to A above. We need to check that $\Sigma\varphi_A$ preserves the operator $+$, and that the diagram above commutes.

▷ Preservation of the operator $+$: let $f, g \in SX$. We have:

$$\begin{aligned} \Sigma\varphi_A(f + g) &= \Sigma\varphi_A(\lambda x. f \ x + g \ x) \\ &= \sum_{x \in X} (f \ x + g \ x) \cdot \varphi_A(x) \\ &= \sum_{x \in X} f \ x \cdot \varphi_A(x) + \sum_{x \in X} g \ x \cdot \varphi_A(x) \\ &= \Sigma\varphi_A(f) + \Sigma\varphi_A(g) \end{aligned}$$

▷ Commutation of the diagram: let $x \in X$. We have:

$$\begin{aligned} (\Sigma\varphi_A)(\eta_{\mathbf{CSG}}(x)) &= \Sigma\varphi_A\left(\lambda y. \begin{cases} 1 & \text{if } x = y \\ 0 & \text{otherwise} \end{cases}\right) \\ &= \sum_{y \in X} \begin{cases} 1 \cdot \varphi_A(x) & \text{if } x = y \\ 0 & \text{otherwise} \end{cases} \\ &= \varphi_A(x) \end{aligned}$$

Therefore, the diagram commutes.

▷ Uniqueness: let $h : F_{\mathbf{CSG}}X \rightarrow A$ be a morphism of **CSG**-models such that the diagram above commutes. We want to show that $h = \Sigma\varphi_A$. Let $f \in SX$. We have:

$$\begin{aligned} h(f) &= h\left(\sum_{x \in X} f \ x \cdot \eta_{\mathbf{CSG}}(x)\right) \\ &= \sum_{x \in X} f \ x \cdot h(\eta_{\mathbf{CSG}}(x)) \\ &= \sum_{x \in X} f \ x \cdot \varphi_A(x) && \text{since the diagram commutes} \\ &= \Sigma\varphi_A(f) \end{aligned}$$

Therefore, we proved that $F_{\mathbf{CSG}}X$ is a free **CSG**-model over X .

Free semigroups

Compared to the free commutative semigroup, we can rely on the order of the terms and the order of applying the operations. We'll use lists to represent the terms of the free semigroup.

Let X be a set. We consider the theory of semigroups **SG**, with a signature Σ with a single operator $(\cdot)$ of arity 2, and with the axiom

$$a, b, c \vdash a \cdot (b \cdot c) = (a \cdot b) \cdot c$$

We pose

$$SX = \{l \in X^* \mid l \text{ is non-empty}\}$$

A DAILY RECAP

- 05/05/2026: first day, talk with Ohad, building tour, admin stuff, setting up the git repo and starting to read the notes of Ohad
- 06/05/2026: reading Ohad's notes, starting to write my own notes. lots of definitions to try and understand how things work.
- 07/05/2026: read the paper on frexes and frals, wrote down the relevant definitions. Meeting with Ohad: 3 equivalent definitions of adjunctions $\Rightarrow$ universal arrows, examples where/when they are used + equivalence of definitions of free algebras and how to construct a free algebra for (commutative) semigroups.

Gave examples of uses of universal arrows: defining adjunctions, defining products and coproducts, and defining free algebras.

Defined and proved the free algebra for commutative semigroups, started to work on the free algebra for semigroups.

B TODO

- dig in the paper about frexes to learn more about that (both frexes and frals) and write the relevant definitions down
- write the definition of free extensions with universal arrows
- compute the free algebras for the semigroup (and commutative semigroup), then combine them and see if this equals the free algebra for the combination (also check that the distributive laws hold)
- look at what was done with involution to take inspiration